STAT/MA 41600

In-Class Problem Set #32 Part 2: November 2, 2026

1. When a certain CEO posts an article on LinkedIN, there are 3 Vice Presidents at her company who usually "Repost" the article. The times until they repost it are X_1, X_2, X_3 (respectively). These are independent exponential random variables with mean 1.5 hours each.

1a. Find $P(X_3 > 2.5 \mid X_3 > 0.75)$, i.e., given that the third person does not repost within the 0.75 hours, find the probability that the third person does not repost within the 2.5 hours.

1b. Let $Y = \min(X_1, X_2, X_3)$. Find the probability density function of Y.

2. Same setup as in Question **1**:

2a. What is the distribution of the number of Vice Presidents who repost the article within 1 hour? What are the parameters of that random variable?

2b. Find the probability mass function of the number of Vice Presidents who repost the article within 1 hour. Please demonstrate that the values of your pmf add up to 1.

3. Same setup as in Question **1**:

3a. Find the CDF of $X_1 + X_2$.

3b. Find the CDF of $X_1 + X_2 + X_3$.

4. Same setup as in Question **1**:

4a. Find the CDF of $\max(X_1, X_2)$.

4b. Find the CDF of $\max(X_1, X_2, X_3)$.

5. Suppose that U is a Continuous Uniform random variable with constant density in the interval $(0, 1)$. Define $X = -\frac{1}{7} \ln U$.

5a. What values can X take on?

5b. What is the CDF of X?

5c. What kind of random variable is X?